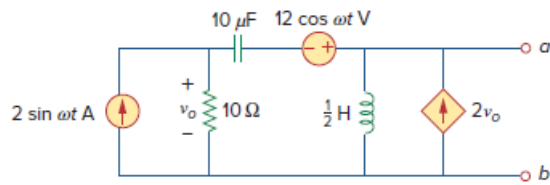


Problem

At terminals a - b , obtain Thevenin and Norton equivalent circuits for the network depicted in Fig. 10.109. Take $\omega = 10$ rad/s.

Figure 10.109:



Step-by-step solution

Step 1 of 8

Refer to the Figure 10.109 in the textbook.

Calculate the impedance of the capacitor.

$$\begin{aligned} Z_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(10)(1 \times 10^{-6})} \\ &= -j10^4 \, \Omega \end{aligned}$$

Calculate the impedance of the inductor.

$$\begin{aligned} Z_L &= j\omega L \\ &= j(10)\left(\frac{1}{2}\right) \\ &= j5 \, \Omega \end{aligned}$$

[Comments \(2\)](#)



Anonymous

shouldn't Z_C be $1/(j*(10)*(10*10^{-6}))$?



Anonymous

nvm

Step 2 of 8

Replace the current source by the open circuit and the voltage source by the short circuit to calculate the Thevenin's impedance, $\mathbf{Z}_{Th}$. Place a test current source to calculate the Thevenin's impedance, $\mathbf{Z}_{Th}$.

Draw the circuit in phasor domain.

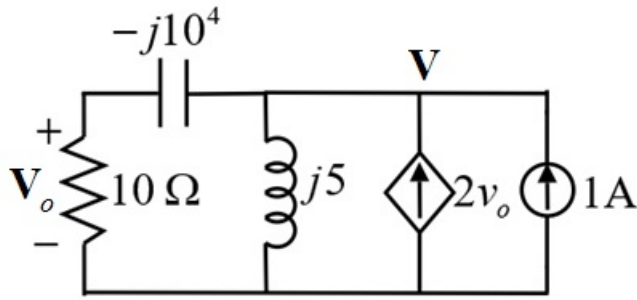


Figure 1

Step 3 of 8

Apply voltage division rule to calculate the voltage, $\mathbf{V}_o$.

$$\mathbf{V}_o = \frac{\mathbf{V} \times 10}{10 - j10^4}$$

Apply the nodal analysis at the node $\mathbf{V}$.

$$1 + 2\mathbf{V}_o = \frac{\mathbf{V}}{j5} + \frac{\mathbf{V}}{10 - j10^4}$$

$$1 + 2\left(\frac{\mathbf{V} \times 10}{10 - j10^4}\right) = \frac{\mathbf{V}}{j5} + \frac{\mathbf{V}}{10 - j10^4}$$

$$\mathbf{V}\left(\frac{19}{10 - j10^4} - \frac{1}{j5}\right) = -1$$

$$\mathbf{V}(1.89 \times 10^{-6} + j1.89 \times 10^{-3} + j0.2) = -1$$

$$\mathbf{V} = \frac{-1}{(1.89 \times 10^{-6} + j0.202)}$$

$$= -4.66 \times 10^{-5} + j4.953$$

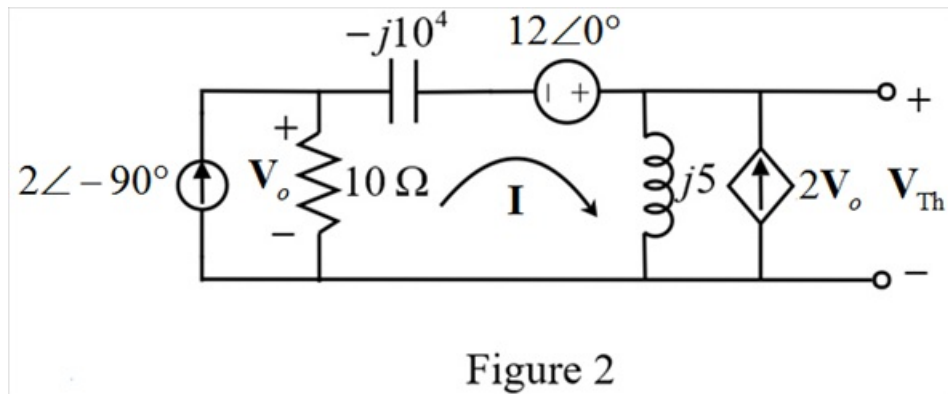
$$= 4.95 \angle 90^\circ \text{ V}$$

Step 4 of 8

Calculate the value of the Thevenin's impedance.

$$\begin{aligned} Z_{Th} &= \frac{V}{I} \\ &= \frac{4.95 \angle 90^\circ}{1} \\ &= j4.95 \, \Omega \end{aligned}$$

Draw the circuit to calculate the Thevenin's voltage.



Step 5 of 8

Apply the Kirchhoff's voltage law in Figure 2.

$$\begin{aligned} (10 - j10^4)I - 10(I - 2\angle -90^\circ) + j5(I - 2V_o) - 12\angle 0^\circ &= 0 \\ (10 - j10^4 + j5)I - 10(-j2) + j10V_o - 12 &= 0 \end{aligned}$$

Substitute $10(2\angle -90^\circ - I)$ for V_o .

$$\begin{aligned} (10 - j10^4 + j5)I - 10(-j2) + j10 \times 10(2\angle -90^\circ - I) - 12 &= 0 \\ (10 - j10^4 + j5)I + j20 + j100(-j2 - I) - 12 &= 0 \\ I(10 - j10^4 + j5 - 100j) + j20 - 12 + 200 &= 0 \\ I &= \frac{-j20 - 188}{(10 - j10^4 - j95)} \end{aligned}$$

$$I = 1.963 \times 10^{-3} - j18.625 \times 10^{-3}$$

[Comments \(1\)](#)



Anonymous

where does the 10 from $(I - 2\angle 90^\circ)$ go

Step 6 of 8

Calculate the value of the Thevenin's voltage.

$$\begin{aligned}
 V_{Th} &= j5(I + 2V_o) \\
 &= j5[I + 2[10(2\angle -90^\circ - I)]] \\
 &= j5[I + [20(-j2 - I)]] \\
 &= j5[I - j40 - 20I] \\
 &= j5[-j40 - 19I]
 \end{aligned}$$

Substitute $1.963 \times 10^{-3} - j18.625 \times 10^{-3}$ for I .

$$\begin{aligned}
 V_{Th} &= j5[-j40 - 19(1.963 \times 10^{-3} - j18.625 \times 10^{-3})] \\
 &= j5(-0.0373 - j39.646) \\
 &= 198.23 - j0.1864 \\
 &= 198.23 \angle -0.054^\circ \text{ V}
 \end{aligned}$$

Step 7 of 8

Draw the Thevenin's equivalent circuit.

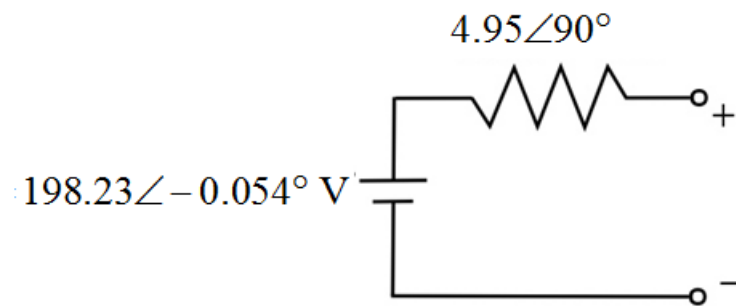


Figure 3

Thus, the Thevenin's equivalent circuit is shown in Figure 3.

Step 8 of 8

Calculate the value of Norton's current.

$$\begin{aligned}
 I_N &= \frac{V_{Th}}{Z_{Th}} \\
 &= \frac{198.23 \angle -0.054^\circ}{4.95 \angle 90^\circ} \\
 &= 40.046 \angle -90.05^\circ \text{ A}
 \end{aligned}$$

Draw the Norton's equivalent circuit.

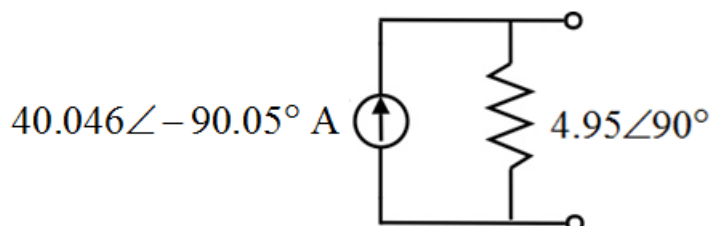


Figure 4

Thus, the Norton's equivalent circuit is shown in Figure 4.